

2 per wagon · cargo temperature (-50...+150°C)



Timing — no polling: this node advertises on its **own** timer — about every **30 s** normally, **~1 s** while in alarm. The gateway's **10 min / 12 h** cadence is a **separate** clock. This node is **never polled**: it broadcasts blind, the gateway's low-power scan (**100 ms every 5 s**, **2% duty**) caches the latest advert, and on the gateway's own schedule it forwards that cache upward.

1. Init: load `node_id`, `wgn_group` and thresholds from NVS; start the coulomb gauge (which auto-zeroes if it detects a replacement cell).
2. Wake on the node's own RTC timer; power sensor rail, read rtd via saadc, sts4x internal temp, bma400 |a|, rail back off.
3. Evaluate against the alarm threshold **with hysteresis** (on > high or < low, off inside band).
4. **Seal** the reading with AES-CCM under the per-wagon key, using a fresh monotonic counter `ctr` as the nonce (persisted in NVS so it never repeats). Nonce byte 7 carries a direction tag so an uplink nonce can never collide with a downlink one.
5. Advertise the encrypted burst (non-connectable); **slow when normal (~30 s), fast while in alarm (~1 s)**. Both periods are read at runtime from `sn_normal_s()` / `sn_fast_s()`, so a `SW_DN_SET_CADENCE` downlink changes them live and the change survives a reboot (section 5.1). A `DEBUG_TRACE` build seeds **10 s** instead of 30 s so a node is visible on the bench without waiting half a minute per advert - the seed only, not a different mechanism.

6. Every CFGWIN_PERIOD_MS of wall-clock time - **including during an alarm** - open a 4 s connectable **config window** so the gateway can write new thresholds, or stream a firmware image (section 6).

3. Advertisement payload (encrypted `sw_adv_enc`, 21 B)

Cleartext header (company, ver, wgn_group, type, id, ctr) + **AES-CCM ciphertext** of the fields below + 4-byte MIC. The header is authenticated; the reading itself is encrypted.

Field	Meaning for this node
wgn_group	This wagon's isolation group, from the wagon number (gateway drops other groups; cleartext)
node_type / node_id	sensor type + unique id within the wagon (cleartext header)
ctr	monotonic nonce counter, persisted in NVS (cleartext; makes each advert unique)
flags / batt (encrypted)	ALARM/LOWBATT flags, battery % from the coulomb gauge
value	RTD cargo temperature $\times 10$ ($^{\circ}\text{C}$) [primary]
value2	STS4x internal temperature $\times 10$ ($^{\circ}\text{C}$)
flags	ALARM (over-temp) IMPACT LOWBATT

3.1 Cadence over the air (`sw_dn_set_cadence`, 0x08)

RDSO §7.15 requires the vendor to set operating values in the field, so the advertising period is not fixed at build time. The command carries the **quiet** period in `value` and the **alarm** period in `hyst`, both in **seconds** - the same two wire fields every other threshold uses, so it inherits the individual / by-type / all-nodes fan-out unchanged.

Guard	Behaviour
Range	Both clamped to <code>CADENCE_MIN_S</code> 1 s ... <code>CADENCE_MAX_S</code> 3600 s. Below 1 s buys nothing over the alarm cadence and costs battery; above an hour the node cannot satisfy the gateway's 15-minute staleness check and would be flagged DOWN continuously.
Ordering	The alarm period is never allowed to exceed the quiet one. An alarm reporting <i>less</i> often than normal traffic would invert the purpose of the fast cadence.
Persistence	Written to NVS (sw/norm, sw/fast) and reloaded by <code>settings_load()</code> , so a cadence tuned over the air survives both a reboot and a firmware update carrying different compile-time defaults.
Effect	Immediate - the very next sleep uses the new value. No reboot.

Battery consequence: the §5.1.20 six-year projection is built on the **30 s** production figure. Dropping the quiet cadence to 5 s multiplies advertising charge by six and invalidates the life estimate - the clamp bounds the damage, it does not prevent it.

4. Alarm / event mapping

Condition	Protocol code	Notes
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Tank/cargo temperature breach	TANK_OVERTEMP	immediate alert on breach
Tamper impact	TANK_TAMPER	off-spec code - not in Protocol Rev.1 §3.2
Cell low	NODE_LOW_BATTERY	advisory

5. Battery gauge (coulomb counting)

The cell is a **non-rechargeable 13 Ah Li-SOCl₂; ER34615**. Its discharge curve is flat, so a voltage-to-percent map reads ~100 % for years and then collapses. State of charge is therefore obtained by **integrating what the firmware spends**: every sleep, measurement, advertising burst and config window adds (current × time) in nAh to an accumulator persisted in NVS. Self-discharge is carried as a constant current so elapsed time is never free.

Item	Value
Usable capacity	BATT_USABLE_MAH = 10000 (derated from the 13 Ah nameplate for cold and load)
Total current ceiling	13 Ah over a 6-year life = 247 µA average
Predicted average	~134 µA (sleep 5 + self-discharge 15 + measure 1 + advertising 60 + config window 53)
Independent end-of-life check	BATT_EOL_MV = 3000 mV, so counter drift cannot hide a dying cell

Calibration outstanding: every BATT_I_*_UA constant except self-discharge is an estimate marked MEASURE: in the source. They must be measured with a Nordic PPK2 in series with the cell over one full wake cycle before the projected life means anything.

6. Firmware update over BLE

The node can be updated by **its own gateway only**. There is no SMP service, no MCUmgr, and no phone path - the image arrives as sealed chunks on a second GATT characteristic during the same config window used for thresholds.

Step	What happens
Stage	Gateway downloads the image over HTTP into its NOR ota partition and computes a CRC16. The two links are never up together, so the image must be staged before any of it can be sent.
Arm	SW_DN_OTA_BEGIN on the config characteristic: erases this node's MCUboot secondary slot and announces the size. Sent once per campaign - a resumed transfer must not repeat it.
Stream	192-byte chunks, each its own AES-CCM frame with nonce direction tag 0x02, accepted only strictly in order . Delivery spans as many windows as it takes; progress is persisted in gateway FRAM so a reset resumes rather than restarts.
Verify	SW_DN_OTA_END carries the CRC16. A mismatch discards the whole transfer before anything is marked bootable.
Swap	boot_request_upgrade(B00T_UPGRADE_TEST). No reboot here - the node swaps on its own next reset, so an update never interrupts a report in flight. MCUboot verifies the fleet signature and auto-reverts if the new image fails to confirm.

Confirm	<code>boot_write_img_confirmed()</code> , called only after the new image completes a full successful cycle - sense, seal, advertise, account. TEST mode without this step reverts on the second reboot , so an update that appeared to install would silently vanish days later. Deferring the confirm until a whole cycle has passed is what makes the auto-revert meaningful: an image that boots but cannot read its sensor is rejected by the node itself.
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Three independent gates, in order: per-chunk AES-CCM (a neighbouring wagon or an attacker cannot inject one byte), CRC16 over the whole image (catches truncation or reordering before anything is bootable), and MCUboot signature verification with auto-revert (the final authority). Gate 3 alone would stop unsigned code running, but not stop a wagon spending hours of radio assembling rubbish - that is what 1 and 2 buy.

During a campaign the gateway sends `SW_DN_FASTWIN` so windows open every few seconds instead of every 10 minutes. Fast mode is **deadlined** in the node, so an abandoned campaign or a lost cancel cannot strand it at high duty.

6. Hardware & pin map (verified against the TANK schematic)

Function	Net	Pin
I2C SDA / SCL (BMA400 + STS4x)	SDA / SCL	P1.04 / P1.05 (i2c21)
Sensor-rail enable (TPS22917)	SW_AXI	P1.06
BMA400 interrupt INT1	BMA_INT	P0.04
RTD analog in (LMV321 bridge amp)	Temp_Sense	P1.07 / AIN3
External watchdog WAKE / DONE (TPL5010)	uC_Wakeup / uC_Alive	P0.00 / P0.01

7. Why this node does NOT keep its sensor rail on

Unlike the DOOR node, the sensor rail here is **gated off between reads**. The RTD bridge is excited by a TL431 reference that draws roughly 1 mA continuously, and it shares `VCC_AXI` with the BMA400. Leaving that rail up to satisfy §7.3 would cost about four times the entire 247 µA budget. The trade is deliberate: continuous tamper watch is provided on the DOOR nodes, and this node powers its rail only to measure.

8. Open items

- **RTD calibration outstanding.** The element type (PT100 vs PT1000) and the LMV321 gain are not determinable from the drawing, so `RTD_CAL_M` / `RTD_CAL_B` must be set from a two-point bench calibration. **Until then the reported cargo temperature is not trustworthy.**
- **STS4x address** 0x44 in firmware vs 0x4A on the pin-map sheet.
- `TANK_TAMPER` is not a Protocol Rev.1 alarm code.